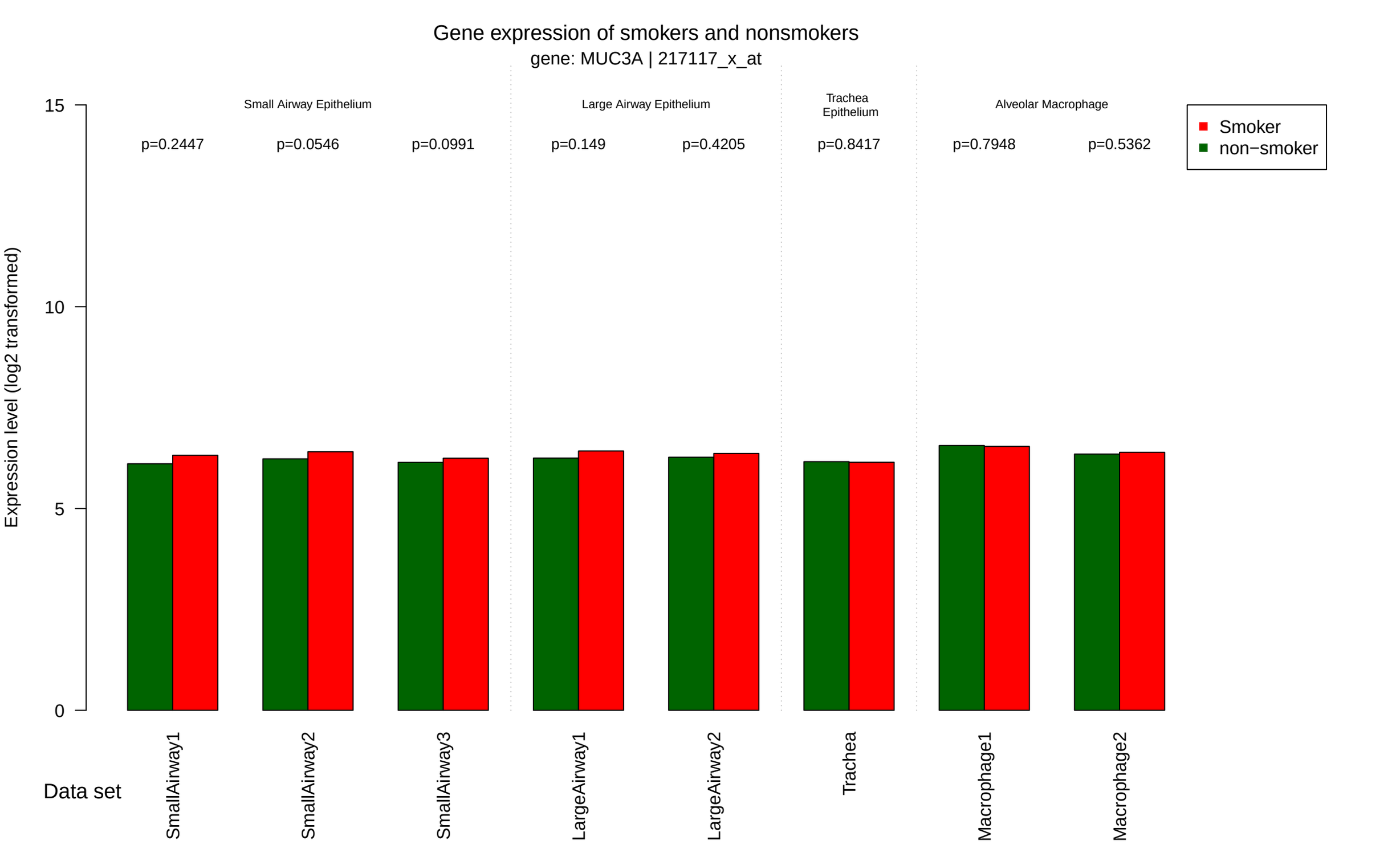
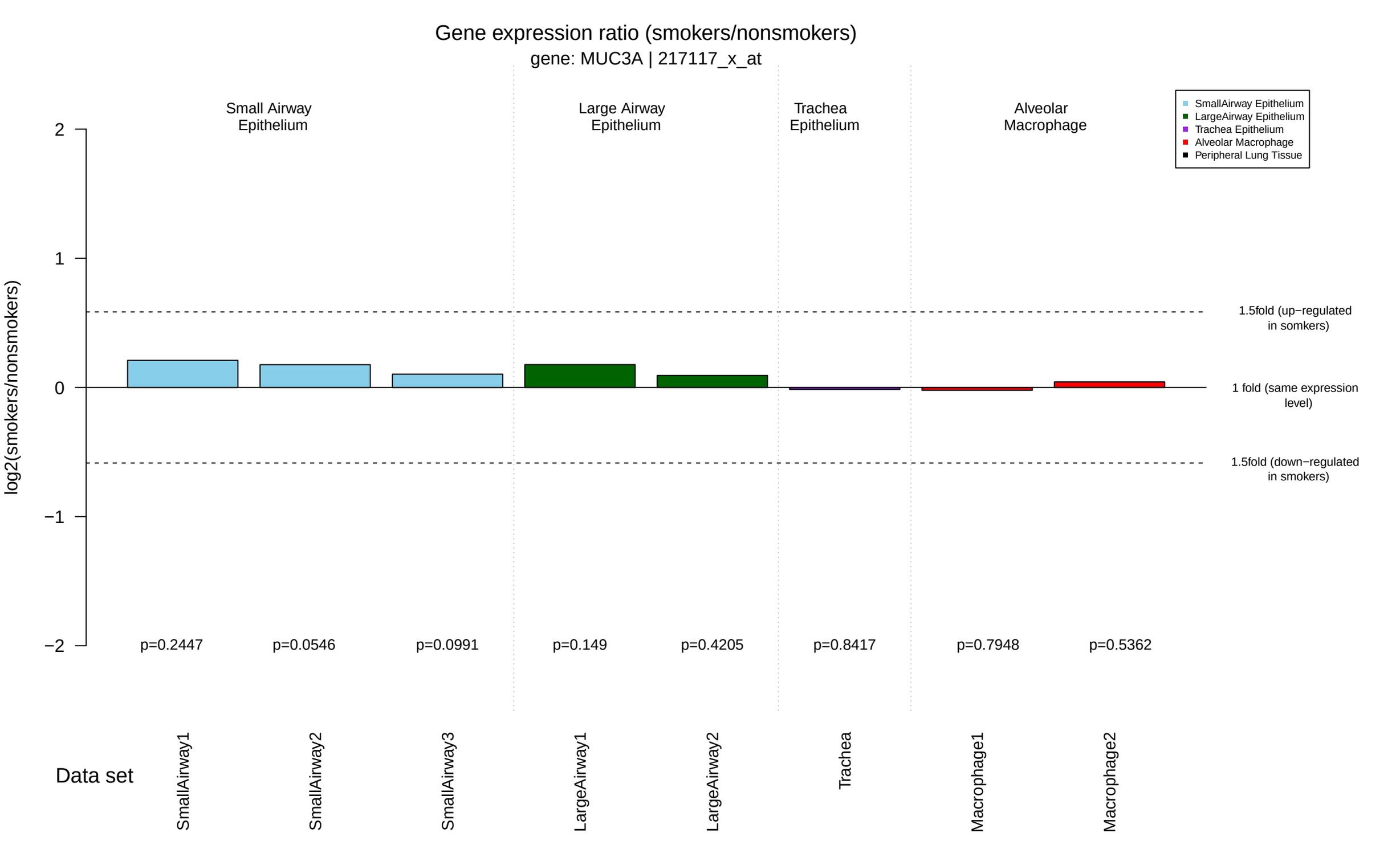
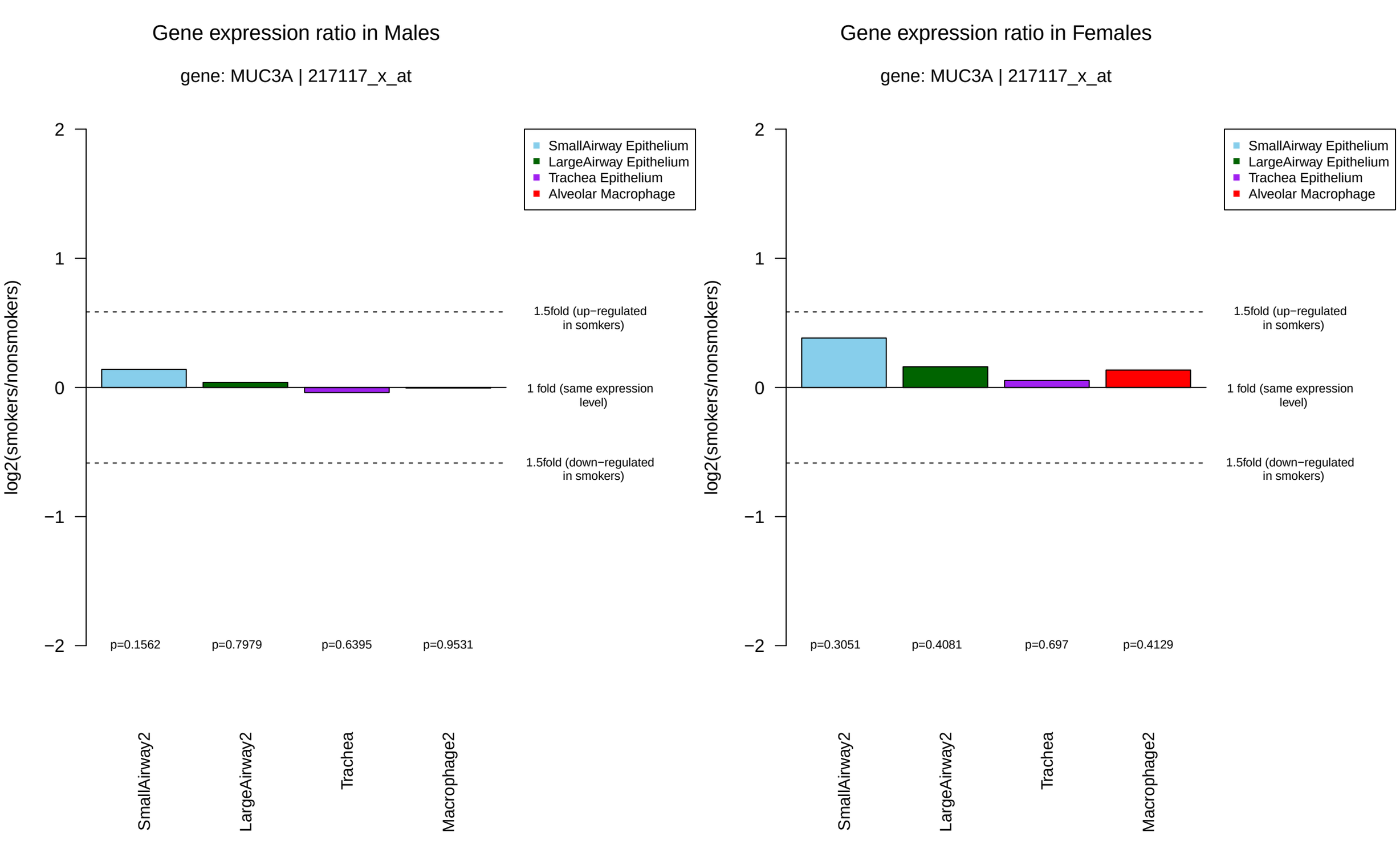
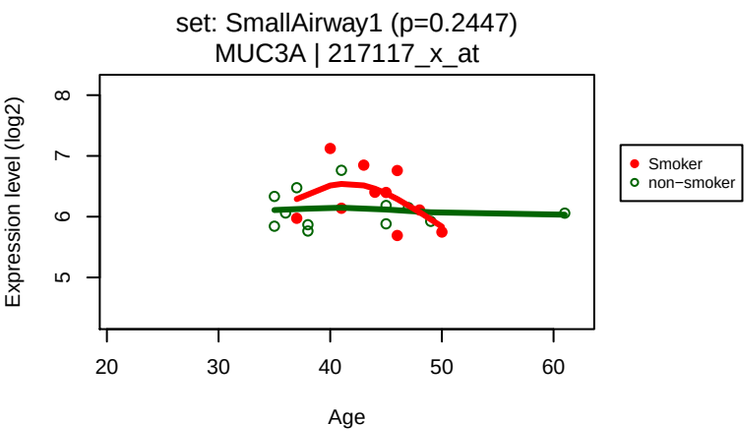
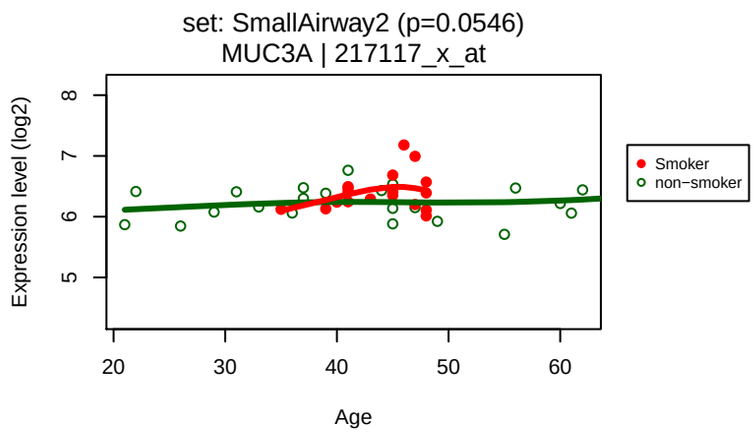
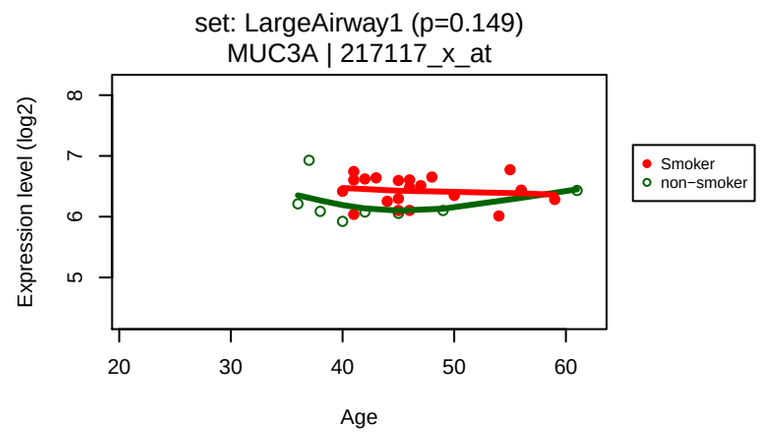

Gene expression of smokers and nonsmokers
gene: MUC3A | 217117_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2447
p=0.0546
p=0.0991
p=0.149
p=0.4205
p=0.8417
p=0.7948
p=0.5362
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

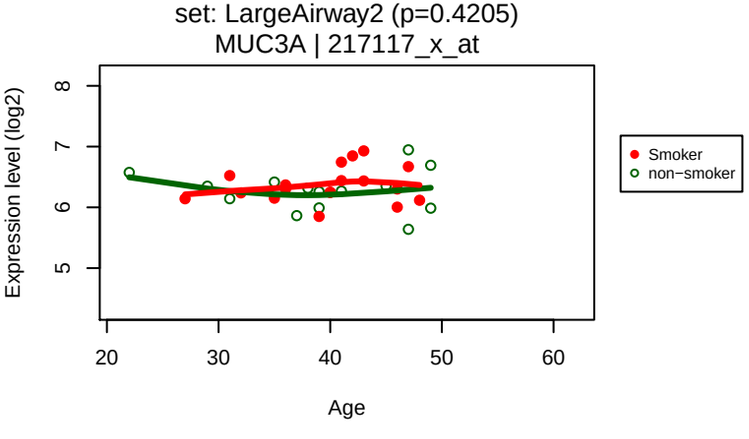

Gene expression ratio (smokers/nonsmokers)
gene: MUC3A | 217117_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2447
p=0.0546
p=0.0991
p=0.149
p=0.4205
p=0.8417
p=0.7948
p=0.5362
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

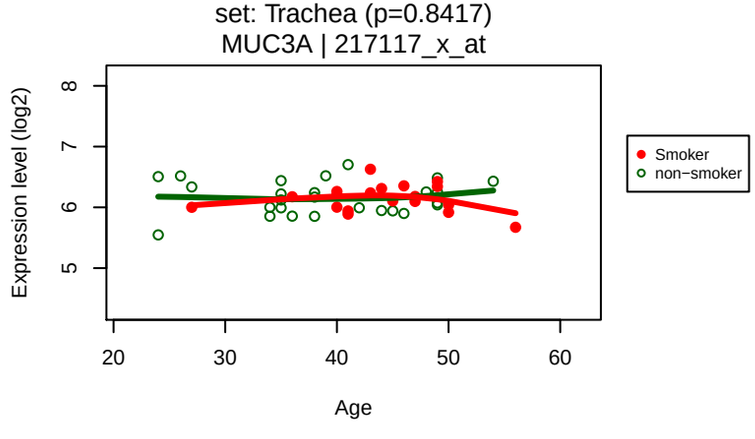

Gene expression ratio in Males
gene: MUC3A | 217117_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1562
p=0.7979
p=0.6395
p=0.9531
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC3A | 217117_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3051
p=0.4081
p=0.697
p=0.4129
SmallAirway2
LargeAirway2
Trachea
Macrophage2

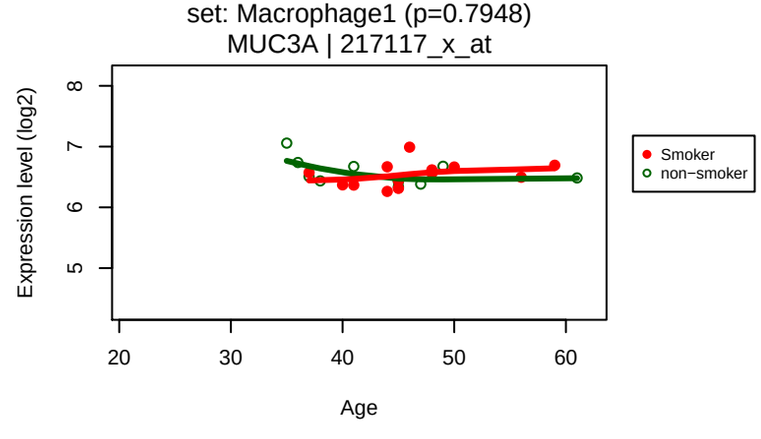

set: SmallAirway1 (p=0.2447)
MUC3A | 217117_x_at
Expression level (log2)
Age
Smoker
non−smoker

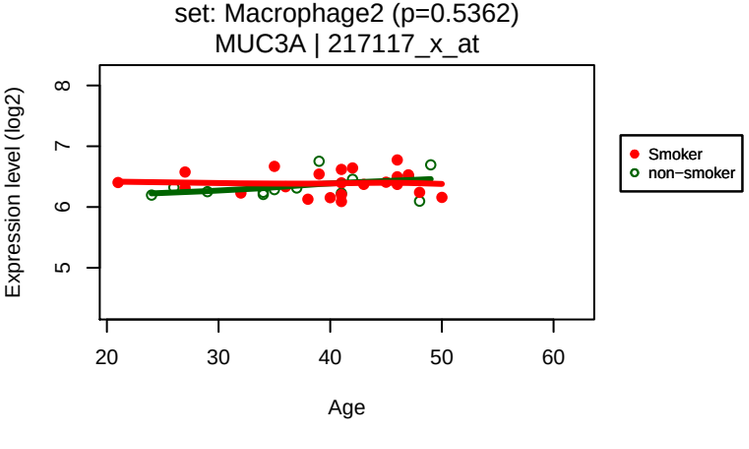

set: SmallAirway2 (p=0.0546)
MUC3A | 217117_x_at
Expression level (log2)
Age
Smoker
non−smoker

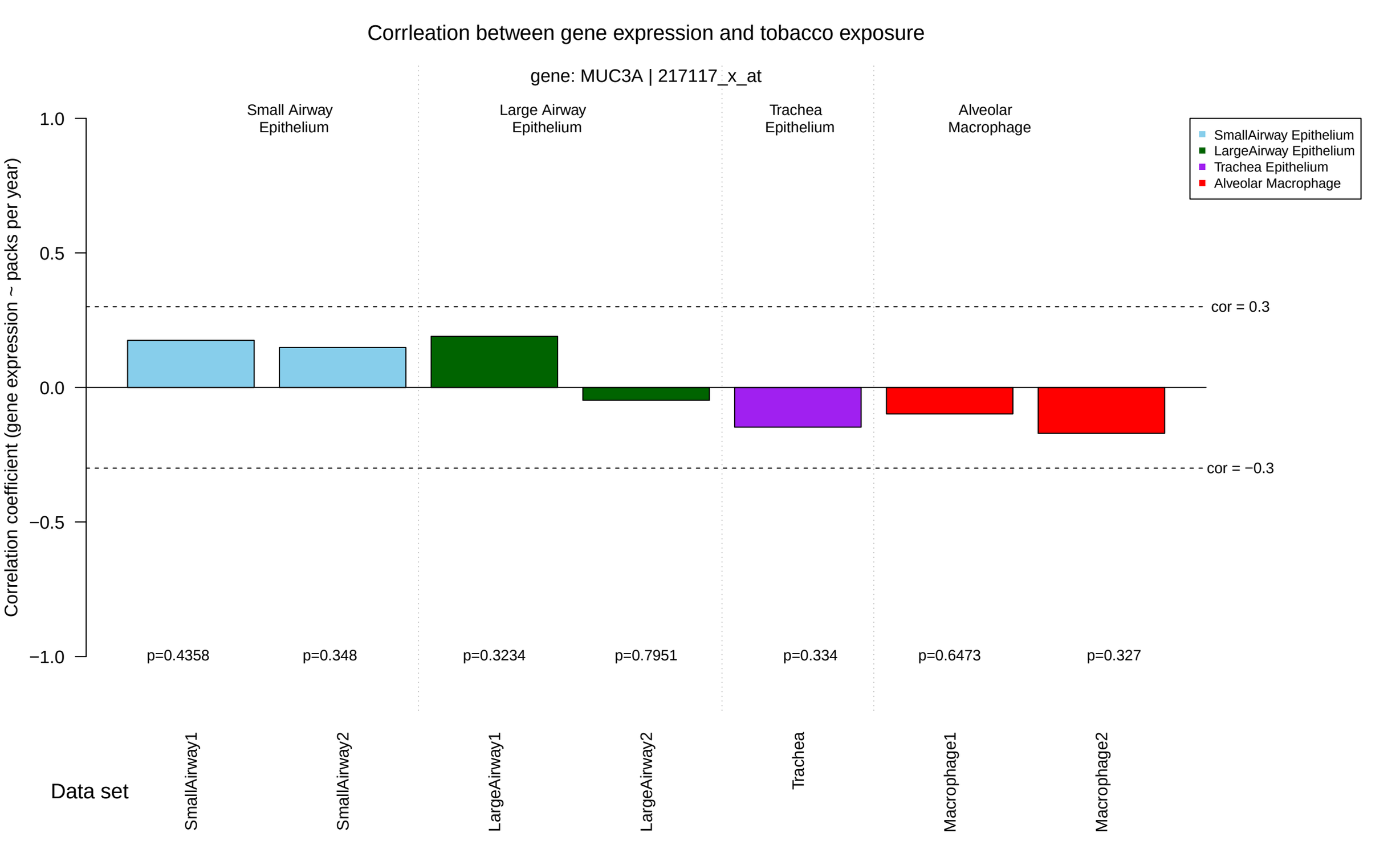

set: LargeAirway1 (p=0.149)
MUC3A | 217117_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4205)
MUC3A | 217117_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8417)
MUC3A | 217117_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7948)
MUC3A | 217117_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5362)
MUC3A | 217117_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC3A | 217117_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4358
p=0.348
p=0.3234
p=0.7951
p=0.334
p=0.6473
p=0.327
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2